Acceptance test book

skeleton-apps on the local stack: Local stack, API with Bruno, Web application, iOS app on the simulator, Android app on the emulator, Command line on macOS

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Tester	
Date of the run	
Machine and macOS	

How to use this book: run the tests in order, section by section. Each test lists its preconditions, the exact steps (commands in grey boxes, to type from the root of the repository unless said otherwise) and the expected result. Tick one box per test: Pass when everything expected happens, Fail when something differs (say what in Notes), Blocked when a precondition could not be met. The buttons and fields can be filled in Preview, Acrobat or any PDF reader with forms: save the file to keep them. The table of contents and the bookmarks lead to each test.

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S. Local stack

Every test runs against the local stack only: PostgreSQL, the mock OpenID Connect provider and the API, with the web app in front of it when the project has one, started by Gradle with Docker Compose. Run the commands from the root of the repository, on the develop branch.

S-01 - Prerequisites are installed

Preconditions

A Mac with Apple Silicon. Nothing of the project is running.

Steps

1. Check Docker with Compose v2 and Git:

```
$ docker compose version
$ git status -sb
```

2. Check the Android SDK variable:

```
$ echo $ANDROID_HOME
```

3. Check Xcode and XcodeGen for the iOS tests:

```
$ xcode-select -p && xcodegen --version
```

Expected result

- docker compose prints a version 2.x or later.
- git shows the develop branch, with no local change.
- ANDROID_HOME points to the Android command-line tools (for example /opt/homebrew/share/android-commandlinetools).
- xcode-select prints /Applications/Xcode.app/Contents/Developer, and xcodegen prints its version.
- Java is not required: Gradle downloads its own JDK (Temurin 25).

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

S-02 - The whole stack starts

Preconditions

S-01 passed. Ports 8088, 8090 and 8091 are free.

Steps

1. Build the images and start the stack:

```
$ ./gradlew startLocalStack
```

Expected result

- The build ends with BUILD SUCCESSFUL.
- The last line reads: Local stack ready: http://localhost:8088 (identity provider: http://localhost:8090).

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

S-03 - Every container runs

Preconditions
S-02 passed.

Steps

```
$ docker compose --profile app ps
```

Expected result
- postgres, mock-oauth2-server and api are listed, running; postgres and api report healthy.
- web is listed too, running.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

S-04 - The API is ready, the identity provider answers

Preconditions
S-02 passed.

Steps

```
$ curl -s http://localhost:8091/actuator/health/readiness
$ curl -s http://localhost:8090/default/.well-known/openid-configuration
```

Expected result
- The readiness probe answers {"status":"UP"}.
- The discovery document is JSON, with "issuer": "http://localhost:8090/default" and an authorization_endpoint and a token_endpoint under that issuer.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

S-05 - No anonymous access

Preconditions
S-02 passed.

Steps

```
$ curl -si http://localhost:8088/api/v1/notes
```

Expected result
- The API answers HTTP 401 with an empty body: no route is open without a token.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

S-06 - The web app is served safely

Preconditions

S-02 passed.

Steps

```
$ curl -sI http://localhost:8088/
```

Expected result

- The web root answers HTTP 200 with Content-Type: text/html.
- The response carries a Content-Security-Policy that includes frame-ancestors 'none' and object-src 'none', and X-Content-Type-Options: nosniff.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B. API with Bruno

The Bruno collection in apps/bruno is generated from the OpenAPI contract and runs as the user alice, with a client credentials token from the mock provider. The curl tests after it check, step by step, what the collection only smokes: versions, validation, ownership and limits. In the commands, B is http://localhost:8088/api/v1.

B-01 - The collection runs green from the command line

Preconditions

The stack is running (S-02).

Steps

```
$ cd apps/bruno
$ npm ci
$ npm run smoke
$ cd ../..
```

Expected result

- The execution summary shows Status PASS.
- Requests: 5 (5 Passed). Tests: 5/5.
- No request answers 401, 403 or a 5xx.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-02 - The collection runs in the Bruno app

Preconditions

The Bruno desktop app is installed. The stack is running.

Steps

1. In Bruno, open the collection folder apps/bruno.
2. Select the environment Stack (top right).
3. Open notes > Write a new note and send it.
4. Open notes > List the caller's notes, most recently updated first and send it.

Expected result

- Bruno fetches a token by itself (OAuth2, client credentials) before the first request.
- Write a new note answers 201 Created, with an ETag header and the note in JSON.
- The list answers 200, with the note in items.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-03 - A token for a test user

Preconditions

The stack is running.

Steps

1. Set the API address, then get a token for alice and one for bob:

```
$ B=http://localhost:8088/api/v1
$ tok() { curl -s -X POST localhost:8090/default/token -d grant_type=client_credentials -d
client_id=$1 -d client_secret=x | python3 -c 'import
sys,json;print(json.load(sys.stdin)["access_token"])'; }
$ ALICE=$(tok alice); BOB=$(tok bob)
$ echo ${ALICE:0:20}
```

Expected result

- Both variables hold a JWT (three dot-separated parts starting with eyJ).
- The token's subject is the client id: alice and bob are two different owners.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-04 - Write a note: 201, Location and ETag

Preconditions

B-03 done in the same terminal.

Steps

```
$ curl -si -X POST $B/notes -H "Authorization: Bearer $ALICE" -H 'Content-Type:
application/json' -d '{"title":"Acceptance","content":"First"}'
```

1. Copy the id of the answer into a variable:

```
$ ID=<the id>
```

Expected result

- HTTP 201, with Location: /api/v1/notes/<id> and ETag: "0".
- The body holds id, title Acceptance, content First, createdAt, updatedAt and version 0.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-05 - Read and list the note

Preconditions

B-04 done.

Steps

```
$ curl -si $B/notes/$ID -H "Authorization: Bearer $ALICE"
$ curl -s "$B/notes?page=0&size=20" -H "Authorization: Bearer $ALICE"
```

Expected result

- The read answers 200 with ETag: "0" and the same note.
- The list answers 200 with items, page 0, size 20, totalItems and hasNext; the note is first: most recently updated first.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

B-06 - A change needs If-Match (428)

Preconditions

B-04 done.

Steps

```
$ curl -s -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H 'Content-Type:
application/json' -d '{"title":"Changed","content":"Second"}' -w ' %{http_code}\n'
```

Expected result

- HTTP 428, a problem+json body with errorCode VERSION_REQUIRED, detail This change needs an If-Match header. and a hint to send the ETag last read.
- The note is unchanged.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

B-07 - A change on the current version succeeds (200, new ETag)

Preconditions

B-06 done.

Steps

```
$ curl -si -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H 'If-Match: "0"' -H '
Content-Type: application/json' -d '{"title":"Changed","content":"Second"}'
```

Expected result

- HTTP 200 with ETag: "1"; the body shows title Changed and version 1.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

B-08 - A change on an outdated version is refused (412)

Preconditions

B-07 done.

Steps

1. Send a change still based on version 0:

```
$ curl -s -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H 'If-Match: "0"' -H '
Content-Type: application/json' -d '{"title":"Stale","content":"x"}' -w ' %{http_code}\n'
$ curl -s $B/notes/$ID -H "Authorization: Bearer $ALICE"
```

Expected result

- HTTP 412, errorCode VERSION_MISMATCH, detail Note <id> is at version 1, not 0.
- Reading the note again still shows title Changed, version 1: nothing was overwritten.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-09 - Invalid input is refused (400)

Preconditions

B-03 done.

Steps

```
$ curl -s -X POST $B/notes -H "Authorization: Bearer $ALICE" -H 'Content-Type:
application/json' -d '{"title":"","content":"x"}' -w ' %{http_code}\n'
$ curl -s -X POST $B/notes -H "Authorization: Bearer $ALICE" -H 'Content-Type:
application/json' -d '{"title":" ","content":"x"}' -w ' %{http_code}\n'
$ curl -s "$B/notes?size=500" -H "Authorization: Bearer $ALICE" -w ' %{http_code}\n'
```

Expected result

- An empty title answers 400, detail Invalid request content. (refused by the contract).
- A blank title answers 400 with errorCode CONSTRAINT_VIOLATION and detail A title must be non-blank and at most 200 characters long (refused by the domain).
- A page of 500 answers 400, detail size must be less than or equal to 100.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

B-10 - Someone else's note does not exist (404)

Preconditions

B-04 done, as alice.

Steps

```
$ curl -s $B/notes/$ID -H "Authorization: Bearer $BOB" -w ' %{http_code}\n'
$ curl -s $B/notes -H "Authorization: Bearer $BOB"
```

Expected result

- bob reading alice's note gets 404, errorCode NOTE_NOT_FOUND: exactly as for a missing note.
- bob's list does not contain alice's note.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

B-11 - A body over 64 KiB is refused (413)

Preconditions

B-03 done.

Steps

```
$ python3 -c 'import json;print(json.dumps({"title":"Big","content":"x"*70000}))' > big.json
$ curl -s -X POST $B/notes -H "Authorization: Bearer $ALICE" -H 'Content-Type:
application/json' --data-binary @big.json -o /dev/null -w '%{http_code}\n'; rm big.json
```

Expected result

- HTTP 413, before the request reaches the API.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

B-12 - Delete needs If-Match, then the note is gone

Preconditions

B-07 done (the note is at version 1).

Steps

```
$ curl -s -X DELETE $B/notes/$ID -H "Authorization: Bearer $ALICE" -w ' %{http_code}\n'
$ curl -s -X DELETE $B/notes/$ID -H "Authorization: Bearer $ALICE" -H 'If-Match: "1"' -w '
%{http_code}\n'
$ curl -s $B/notes/$ID -H "Authorization: Bearer $ALICE" -w ' %{http_code}\n'
```

Expected result

- Without If-Match: 428 VERSION_REQUIRED, and the note stays.
- With If-Match "1": 204 No Content.
- Reading it again: 404 NOTE_NOT_FOUND.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

W. Web application

The Angular app is served by nginx on http://localhost:8088. Use a desktop browser (Chrome, Firefox or Safari), ideally in a private window. The mock provider accepts any username: the username becomes the owner of the notes, so alice in the browser is the same owner as alice in the API tests.

W-01 - Opening the app asks to sign in

Preconditions

The stack is running. No session in the browser.

Steps

1. Open http://localhost:8088.

Expected result

- The browser is sent to the provider's page on localhost:8090 (address starting with http://localhost:8090/default/authorize).
- The page shows a username field and a Sign-in button.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-02 - Sign in and see your notes

Preconditions

W-01 done.

Steps

1. Type alice as username, leave the other field empty, then click Sign-in.

Expected result

- The browser comes back to http://localhost:8088.
- The page shows the New note form (Title, Content, Add note) and Your notes (N), with alice's notes, or No notes yet.
- A Sign out button is visible.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-03 - A note needs a title

Preconditions
Signed in as alice.

Steps
1. Leave Title empty and click Add note.

Expected result
- The message A title is required, up to 200 characters. appears under Title.
- No note is created: the count of Your notes does not change.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

W-04 - Write a note

Preconditions
Signed in as alice.

Steps
1. Title: Groceries. Content: Milk, eggs. Click Add note.

Expected result
- The note appears first under Your notes, with its title and content.
- The count goes up by one; the form is empty again.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

W-05 - Edit a note, or cancel

Preconditions
W-04 done.

Steps
1. On Groceries, click Edit. The heading becomes Edit note and the form holds the note.
2. Change the title to Weekend groceries, click Save.
3. Click Edit again, change the title, then click Cancel.

Expected result
- After Save, the note shows Weekend groceries and moves first.
- After Cancel, the note keeps Weekend groceries and the form is back to New note.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

W-06 - A note changed elsewhere is not overwritten

Preconditions

Signed in as alice, Weekend groceries exists.

Steps

1. Open http://localhost:8088 in a second tab (same session).
2. In tab 1, click Edit on Weekend groceries.
3. In tab 2, edit the same note, set the title to From tab 2, Save.
4. In tab 1, set the title to From tab 1, Save.

Expected result

- Tab 1 shows the alert This note was changed elsewhere. It now shows the latest version.
- The form of tab 1 now holds From tab 2: tab 1's change was not saved.
- After a reload, both tabs list From tab 2.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-07 - Delete only once confirmed

Preconditions

Signed in as alice, at least one note.

Steps

1. Click Delete on a note: Delete this note? appears with Delete and Keep.
2. Click Keep. Then Delete again, and confirm with Delete.

Expected result

- After Keep, the note stays.
- After Delete, the note leaves the list and the count goes down by one.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-08 - Notes come ten per page

Preconditions

Signed in as alice. B-03 done in a terminal (ALICE token).

Steps

1. Create twelve notes for alice from the terminal:

```
$ for i in $(seq 1 12); do curl -s -o /dev/null -X POST $B/notes -H "Authorization: Bearer
$ALICE" -H 'Content-Type: application/json' -d "{\"title\":\"Page note
$i\",\"content\":\"\"}"; done
```

2. Reload the web app, then use Next and Previous.

Expected result

- The first page lists ten notes, Page note 12 first; Previous is disabled.
- Next shows the remaining notes; on the last page Next is disabled.
- The count in Your notes (N) is the total, not the page size.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-09 - Each user sees only their notes

Preconditions

Signed in as alice with notes.

Steps

1. Click Sign out, open http://localhost:8088 again, sign in as bob.

Expected result

- bob sees none of alice's notes (No notes yet., unless bob wrote some in B-10).
- Signing back in as alice shows her notes again.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-10 - Sign out ends the session

Preconditions

Signed in.

Steps

1. Click Sign out, then open http://localhost:8088 again.

Expected result

- Sign out leads back to the provider's sign-in page.
- Opening the app again asks to sign in: the session is over.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

W-11 - No error in the browser console

Preconditions

Signed in, the browser's developer tools open on Console.

Steps

1. Repeat W-04 to W-07 while watching the console.

Expected result

- No error and no Content-Security-Policy violation is logged.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

I. iOS app on the simulator

The SwiftUI app runs on the iPhone 17 simulator and reaches the stack on localhost directly: the simulator shares the Mac's network. Gradle generates the Xcode project, builds the shared Kotlin module and the app, and launches it. To simulate an outage, the tests stop the web container, which serves the API on 8088, then start it again. Keep a terminal at the root of the repository with B-03's variables, for the API checks.

I-01 - The app builds and launches

Preconditions

The stack is running. Xcode, XcodeGen and the iPhone 17 simulator are installed.

Steps

```
$ ./gradlew iosRun
```

1. If several runtimes have an iPhone 17, pass its identifier:

```
$ xcrun simctl list devices available | grep 'iPhone 17'
$ ./gradlew iosRun -PiosSimulator=<device identifier>
```

Expected result

- The build ends with BUILD SUCCESSFUL; the simulator boots.
- The Notes app opens on its sign-in screen with a Sign in button.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

I-02 - Cancelling the sign-in is handled

Preconditions

I-01 done.

Steps

1. Tap Sign in, then Cancel in the system dialog (or close the sign-in sheet).

Expected result

- The app stays on the sign-in screen and shows Sign-in cancelled.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

I-03 - Sign in and see your notes

Preconditions

I-01 done.

Steps

1. Tap Sign in. iOS asks whether Notes may use localhost to sign in: tap Continue.
2. On the provider's page, type alice and tap Sign-in.

Expected result

- Back in the app, the list is titled Your notes (N) and shows alice's notes: the same titles as the API lists for alice.
- The toolbar shows Sign out and New note.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

I-04 - Write, edit and delete a note

Preconditions

Signed in as alice.

Steps

1. Tap New note. Title: From iPhone. Content: Simulator. Tap Add note.
2. Tap the note, change the title to From iPhone, edited, tap Save.
3. Check what the API holds:

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import
sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

4. Open the note again, tap Delete; Delete this note? appears: tap Keep, then Delete again and confirm.

Expected result

- Add note returns to the list, with the note first; the count goes up.
- Save shows the new title in the list, and the API lists From iPhone, edited.
- Keep leaves the note; the confirmed Delete removes it from the list.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

I-05 - An empty title is refused without calling the API

Preconditions

Signed in.

Steps

1. Tap New note, leave Title empty, tap Add note.

Expected result

- The editor shows title: must not be empty and stays open; no note is created.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

I-06 - Offline: notes kept on the device, changes queued

Preconditions

Signed in, the list shows notes.

Steps

```
$ docker compose stop web
```

1. In the app, tap New note, Title: Written offline, tap Add note.

Expected result

- The list shows Offline: these are the notes kept on this device. with a Retry button.
- The notes read before are still listed; Written offline appears labelled Not synced.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

I-07 - Back online: the queued change reaches the API

Preconditions

I-06 done.

Steps

```
$ docker compose start web
```

1. Wait about five seconds, then tap Retry in the app.

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import
sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- The Offline line and the Not synced label disappear.
- The API lists Written offline.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

I-08 - A conflict is shown and resolved

Preconditions

Signed in as alice; the note Written offline exists and is synced (I-07).

Steps

```
$ docker compose stop web
```

1. On the iPhone, change Written offline to Mine (iPhone), Save: it is labelled Not synced.

```
$ docker compose start web
```

2. Wait about five seconds.
3. Change the note through the API, as another device would (B-03 done in this terminal):

```
$ ID=$(curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(next(n["id"] for n in json.load(sys.stdin)["items"] if n["title"]==sys.argv[1]))' "Written offline")
$ V=$(curl -s $B/notes/$ID -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(json.load(sys.stdin)["version"])')
$ curl -s -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H "If-Match: \"$V\"" -H 'Content-Type: application/json' -d '{"title":"Theirs (API)","content":""}' -o /dev/null -w '%{http_code}\n'
```

4. On the iPhone, tap Retry, then open the note.
5. Tap Keep both.

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- The change through the API answers 200.
- After Retry, the note is labelled Conflict: open to resolve.
- The note shows This note was changed elsewhere., then Yours (Mine (iPhone)) and Theirs (Theirs (API)), with Keep mine, Keep theirs and Keep both.
- Keep both leaves two notes, Mine (iPhone) and Theirs (API), on the iPhone and in the API's list.
- Optional: repeat with Keep mine (the API then holds Mine) and with Keep theirs (the iPhone then shows Theirs).

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

I-09 - The session survives a relaunch

Preconditions

Signed in.

Steps

1. Quit the app from the app switcher, then open Notes again.

Expected result

- The app opens on Your notes without asking to sign in.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

I-10 - Signing out with changes not synced asks first

Preconditions

Signed in.

Steps

```
$ docker compose stop web
```

1. Add a note, then tap Sign out.
2. Tap Stay. Tap Sign out again, then Sign out anyway.

```
$ docker compose start web
```

Expected result

- An alert reads Changes not synced yet: 1. Signing out loses them., with Sign out anyway and Stay.
- Stay keeps the session; Sign out anyway leads to the sign-in screen.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A. Android app on the emulator

The Compose app runs on an Android emulator. adb reverse makes the emulator's localhost the Mac's, so the tokens carry the issuer the API checks; 10.0.2.2 would not work. To simulate an outage, the tests stop the web container, which serves the API on 8088, then start it again. Keep a terminal at the root of the repository with B-03's variables.

A-01 - An emulator is available

Preconditions

ANDROID_HOME is set (S-01).

Steps

1. Once, install a system image and create a device:

```
$ sdkmanager --sdk_root="$ANDROID_HOME" "emulator" "
system-images;android-36;google_apis_playstore;arm64-v8a"
$ avdmanager create avd -n notes -k "system-images;android-36;google_apis_playstore;arm64-v8a"
 -d pixel_7
```

2. Every time, start it:

```
$ emulator -avd notes &
$ adb devices
```

Expected result

- The emulator window opens on the Android home screen.
- adb devices lists one device, emulator-5554, as device.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A-02 - The app installs

Preconditions

A-01 done, the stack is running.

Steps

```
$ adb reverse tcp:8088 tcp:8088
$ adb reverse tcp:8090 tcp:8090
$ ./gradlew :apps:clients:android:installDebug
```

Expected result

- Each adb reverse prints the port (8088, then 8090).
- The build ends with BUILD SUCCESSFUL and Installed on 1 device.
- Notes appears in the emulator's app list.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A-03 - Cancelling the sign-in is handled

Preconditions
A-02 done.

Steps
1. Open Notes, tap Sign in, then close the browser tab with its X (or the back button).

Expected result
- The app shows Sign-in cancelled. and stays on the sign-in screen.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A-04 - Sign in and see your notes

Preconditions
A-02 done.

Steps
1. Tap Sign in. The first time, Chrome may show its welcome screen: choose to use it without an account.
2. On the provider's page, type alice and tap Sign-in.

Expected result
- Back in the app, the top bar reads Your notes (N), with alice's notes, and a Sign out action.
- A New note button is shown at the bottom right.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A-05 - Write, edit and delete a note

Preconditions
Signed in as alice.

Steps
1. Tap New note. Title: From Android. Content: Emulator. Tap Add note.
2. Tap the note, change the title to From Android, edited, tap Save.
3. Check what the API holds:

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import
sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

4. Open the note again, tap Delete; Delete this note? appears: tap Keep, then Delete and confirm with Delete.

Expected result
- Add note returns to the list with the note first; the count goes up.
- Save shows the new title in the list, and the API lists From Android, edited.
- Keep leaves the note; the confirmed Delete removes it.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

A-06 - An empty title is refused without calling the API

Preconditions

Signed in.

Steps

1. Tap New note, leave Title empty, tap Add note.

Expected result

- The editor shows title: must not be empty and stays open; no note is created.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

A-07 - Offline, then back online

Preconditions

Signed in, the list shows notes. If I-06 left a note titled Written offline, delete it first.

Steps

```
$ docker compose stop web
```

1. Tap New note, Title: Written offline, tap Add note.

```
$ docker compose start web
```

2. Wait about five seconds, then tap Retry.

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import
sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- While offline: Offline: these are the notes kept on this device. with Retry; the new note is labelled Not synced.
- After Retry: the offline line and the label disappear; the API lists Written offline.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

A-08 - A conflict is shown and resolved

Preconditions

Signed in as alice; Written offline exists and is synced (A-07).

Steps

```
$ docker compose stop web
```

1. On Android, change the note to Mine (Android), Save: labelled Not synced.

```
$ docker compose start web
```

2. Wait about five seconds.
3. Change the note through the API, as another device would (B-03 done in this terminal):

```
$ ID=$(curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(next(n["id"] for n in json.load(sys.stdin)["items"] if n["title"]==sys.argv[1]))' "Written offline")
$ V=$(curl -s $B/notes/$ID -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(json.load(sys.stdin)["version"])')
$ curl -s -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H "If-Match: \"$V\"" -H 'Content-Type: application/json' -d '{"title":"Theirs (API)","content":""}' -o /dev/null -w '%{http_code}\n'
```

4. On Android, tap Retry, open the note, tap Keep mine.

```
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- The change through the API answers 200.
- After Retry, the note is labelled Conflict: open to resolve.
- The note shows This note was changed elsewhere., Yours and Theirs, and Keep mine, Keep theirs, Keep both.
- Keep mine: Android and the API both hold Mine (Android).

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

A-09 - Signing out with changes not synced asks first

Preconditions

Signed in.

Steps

```
$ docker compose stop web
```

1. Add a note, then tap Sign out.
2. Tap Stay, then Sign out again and Sign out anyway.

```
$ docker compose start web
```

Expected result

- A dialog reads Changes not synced yet: 1. Signing out loses them., with Sign out anyway and Stay.
- Stay keeps the session; Sign out anyway leads back to Sign in.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C. Command line on macOS

notes is a Kotlin/Native binary. Against the local stack, the provider has no device flow, so notes login opens the sign-in page in the browser and waits for its redirect on 127.0.0.1:8770. Tokens go to the login Keychain (item dev.ylp.notes), the notes and pending changes to ~/.local/share/notes. In the commands, notes is the alias of C-01. To simulate an outage, the tests stop the web container, which serves the API on 8088, then start it again.

C-01 - The binary builds

Preconditions

At the root of the repository; the stack may be stopped.

Steps

```
$ ./gradlew :apps:clients:cli:linkReleaseExecutableMacosArm64
$ alias notes="$PWD/apps/clients/cli/build/bin/macosArm64/releaseExecutable/notes.kexe"
$ notes --help
```

Expected result

- BUILD SUCCESSFUL, and the file notes.kexe exists.
- The help lists login, logout, list, show, add, edit, delete, sync and resolve.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-02 - Without a session, a command asks to sign in

Preconditions

The stack is running. Start clean: notes logout --force.

Steps

```
$ notes list; echo "exit $?"
```

Expected result

- Not signed in: run notes login. and exit 1.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-03 - Sign in in the browser

Preconditions

C-02 done.

Steps

```
$ notes login
```

1. The browser opens the provider's page (the terminal also prints its address). Type alice, click Sign-in.

```
$ security find-generic-password -s dev.ylp.notes -a tokens > /dev/null && echo stored
```

Expected result

- The browser shows Signed in. You can close this tab and go back to the terminal.
- The terminal prints Signed in.; nothing else is logged.
- The Keychain check prints stored. The tokens never appear in a command line.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-04 - List, add, show, edit

Preconditions

Signed in as alice.

Steps

```
$ notes list
$ notes add --title "From the CLI" --content "Terminal"
$ notes show <id printed by add>
$ notes edit <id> --title "From the CLI, edited"
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import
sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- list prints Your notes (N), then one line per note: id and title, alice's notes from every client.
- add prints Added <id>.; show prints the title, the content and id <id>, version 0.
- edit prints Saved.; the API lists From the CLI, edited.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-05 - Delete asks first

Preconditions

C-04 done.

Steps

```
$ notes delete <id> (answer n)
$ notes delete <id> (answer y)
```

Expected result

- The CLI asks Delete "From the CLI, edited"? and waits; answering n prints Kept. and the note stays.
- y prints Deleted. and the note leaves notes list. --yes skips the question.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-06 - Offline, then sync

Preconditions

Signed in. If an earlier section left a note titled Written offline, delete it first.

Steps

```
$ docker compose stop web
$ notes add --title "Written offline"
$ notes list
$ notes sync; echo "exit $?"
$ docker compose start web
```

1. Wait about five seconds.

```
$ notes sync
```

Expected result

- add prints Added <id>. Not synced yet: it leaves once the service can be reached.
- list starts with Offline: these are the notes kept on this computer. and marks the note [not synced].
- The first sync fails: The service cannot be reached: changes not synced yet: 1. and exit 1.
- Once the service is back, sync prints Synced.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

C-07 - A conflict is shown and resolved

Preconditions

Signed in; Written offline exists and is synced (C-06).

Steps

```
$ docker compose stop web
$ notes edit <id> --title "Mine (CLI)"
$ docker compose start web
```

1. Wait about five seconds.
2. Change the note through the API, as another device would (B-03 done in this terminal):

```
$ ID=$(curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(next(n["id"] for n in json.load(sys.stdin)["items"] if n["title"]==sys.argv[1]))' "Written offline")
$ V=$(curl -s $B/notes/$ID -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;print(json.load(sys.stdin)["version"])')
$ curl -s -X PUT $B/notes/$ID -H "Authorization: Bearer $ALICE" -H "If-Match: \"$V\"" -H 'Content-Type: application/json' -d '{"title":"Theirs (API)","content":""}' -o /dev/null -w '%{http_code}\n'
$ notes sync
$ notes list
$ notes show <id>
$ notes resolve <id> both
$ curl -s "$B/notes?size=100" -H "Authorization: Bearer $ALICE" | python3 -c 'import sys,json;[print(n["title"]) for n in json.load(sys.stdin)["items"]]'
```

Expected result

- The change through the API answers 200.
- sync prints Synced. Changes in conflict: 1, see notes list.
- list marks the note [conflict].
- show prints This note was changed elsewhere., Yours: Mine (CLI), Theirs: Theirs (API), and Resolve with: notes resolve <id> <mine, theirs, both>.
- resolve prints Resolved.; the API lists both Mine (CLI) and Theirs (API).

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

C-08 - Sign out keeps changes not synced unless forced

Preconditions

Signed in.

Steps

```
$ docker compose stop web
$ notes add --title "Pending"
$ notes logout; echo "exit $?"
$ notes logout --force
$ security find-generic-password -s dev.ylp.notes -a tokens > /dev/null || echo gone
$ docker compose start web
```

Expected result

- logout refuses: Changes not synced yet: 1. Signing out loses them: run notes logout --force to do it anyway. and exit 1.
- logout --force prints Signed out.
- The Keychain check prints gone: the tokens are removed.

Result ☐ Pass ☐ Fail ☐ Blocked

Notes

T. Teardown

Stop what the tests started and leave the machine clean.

T-01 - Everything stops cleanly

Preconditions

All other tests done.

Steps

1. Stop the emulator and the simulator:

```
$ adb emu kill
$ xcrun simctl shutdown all
```

2. Stop the stack:

```
$ ./gradlew stopLocalStack
```

3. To also delete the test data: ./gradlew resetDatabase

Expected result

- stopLocalStack ends with BUILD SUCCESSFUL; docker compose --profile app ps lists nothing.
- The database volume is kept unless resetDatabase was run.

Result	☐ Pass	☐ Fail	☐ Blocked
Notes			

Summary and sign-off

Count the results of each section, then record the verdict.

Section	Tests	Passed	Failed	Blocked
S - Local stack	6			
B - API with Bruno	12			
W - Web application	11			
I - iOS app on the simulator	10			
A - Android app on the emulator	9			
C - Command line on macOS	8			
T - Teardown	1			
Total	**57**			

Verdict

Accepted, accepted with reservations, or rejected; list the failed tests that motivate it.

Reservations

Name

Date